ME 5309 – Mid-Semester Examination: ANSWER SHEET

Roll No.: ME25M019 **Name:** Manish Yadav

Questions attempted: Q1, Q2, Q3

Question 1

(a) (i) $f_x = f/p = 16\times10^{-3} / 5\times10^{-6} =$ **3200 px**.

(ii) Sensor width = 1280 × 5 µm = 6.4 mm. Using similar triangles, FOV = $W_s \cdot f / Z = 6.4 \times 16/400 =$ **0.256 mm**.

Distance per pixel = FOV/1280 = 0.256/1280 = **2×10^{-4} mm** = 0.2 µm.

(b) LSB = $V_{range}/2^n = 5\ V / 2^{12} = 5/4096 = 1.2207$ mV.

Smallest resolvable force ΔF = LSB / S = 1.2207 mV / (10 mV/N) = **0.1221 N**.

(c) Using forward Euler: $x_{k+1} = x_k + T(A\,x_k + B\,u_k)$, so F = 1 + AT and G = BT.

F = 1 + (−2)(0.1) = **0.8**, G = 1 × 0.1 = **0.1**.

$x_{k+1} = 0.8\,x_k + 0.1\,u_k$.

Question 2

(a) $x_1 = \theta$, $x_2 = \theta^{\cdot}$. $x^{\cdot}_1 = x_2$, $x^{\cdot}_2 = (1/J)(\tau - b\,x_2 - mg\ell \sin x_1)$.

Equilibrum means $x^{\cdot} = 0$, so $x_2 = 0$ and $mg\ell \sin\theta_e = \tau_0$. Sence sine is periodic we can always find a θ_e that satisfies this, so an equilibrium exists for any τ_0.

(b) Linearising: A = [0, 1; $\partial f_2/\partial x_1$, $\partial f_2/\partial x_2$].

$\partial f_2/\partial x_1 = -(mg\ell/J)\sin\theta_e = -122.625 \times 0.8660 = -106.2$. $\partial f_2/\partial x_2 = -b/J = -2.5$.

A = [0, 1; −106.2, −2.5], B = [0; 1/J] = **[0; 50]**.

(c) At rest $\theta^{\cdot} = \theta^{\cdot\cdot} = 0$, so $K_p(\theta_d - \theta_{ss}) = mg\ell \sin\theta_{ss}$, i.e. $50(90 - \theta_{ss}) = 2.4525 \sin\theta_{ss}$.

Graviy torqe is non-zero at 90°, so the left side can't be zero, hence $\theta_{ss} \neq \theta_d$.

Taking $\sin\theta_{ss} \approx 1$: $90 - \theta_{ss} = 2.4525/50 = 0.049$, so the steady-state error ≈ **0.049°**.

(d) Error: $K_p(\theta_d - \theta_{ss}) = 0$ at rest, so $\theta_{ss} = \theta_d$; **error = 0**.

Poles: yes they change. The graviy term is removed from the dynamics so the effective stiffness changes from $K_p + mg\ell\cos\theta$ to K_p, which moves the poles (the system becomes faster).

Question 3

(a) (1) Diferentiation is a high-pass operation (gain ∝ ω), so encoder measurement/quantisaton noice is amplifed, by about 1/T = 1000 at 1 kHz.

(2) Encoder quantisation: Δθ is a whole number of counts, so $\hat{\omega} = \Delta\theta/T$ moves in steps of (count size)/T; at low speed the estimate is a coarse staircase of zeros and spikes.

(b) $\omega_c = 20$ rad/s. Tustin with T = 0.001: $a = \omega_c T/2 = 0.01$.

$y_k = [a/(1 + a)](u_k + u_{k-1}) + [(1 - a)/(1 + a)]\,y_{k-1} =$ **$0.00990\,(u_k + u_{k-1}) + 0.98020\,y_{k-1}$**.

DC gain: 2(0.0099)/(1 − 0.9802) = 0.0198/0.0198 = 1 ✓.

(c) A − LC = $[-\ell_1, 1; -\ell_2, -2]$. $\det(sI - A + LC) = (s + \ell_1)(s + 2) - \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 - \ell_2)$.

Desired $(s + 10)^2 = s^2 + 20s + 100$. So $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$ and $2\ell_1 - \ell_2 = 100 \Rightarrow \ell_2 = 36 - 100 = -64$.

$L = [18, -64]^T$.

(d) Advantage of the obsever: it uses the model and the input u, so there is no lag in the estimat, wich a filter always introduces.

The filter is safer when the sampling rate is very high, because the observer would not be able to run fast enough.